THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE UNIT TEST SET 1

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A student wants to find out how oil temperature affects the puffing of a puri. The factor she deliberately changes is called the: 1

(a) constant **(b)** dependent variable
(c) conclusion **(d)** independent variable

2. Who first observed and named 'cells' in thin slices of cork in 1665? 1

(a) Louis Pasteur **(b)** Robert Brown
(c) Robert Hooke **(d)** Anton van Leeuwenhoek

3. According to the World Health Organization, health is a state of complete: 1

(a) wealth and comfort **(b)** physical strength only
(c) absence of fever only **(d)** physical, mental and social well-being

4. A device that provides the electrical energy to drive a current in a circuit is a: 1

(a) cell or battery **(b)** switch **(c)** wire **(d)** bulb

5. A scientific investigation usually begins with: 1

(a) a question or curiosity about an observation
(b) writing a final report
(c) buying equipment
(d) announcing the result

6. The correct order of increasing complexity is: 1

(a) tissue, cell, organ, organism
(b) cell, organ, tissue, organism
(c) organ, tissue, cell, organism
(d) cell, tissue, organ, organ system, organism

7. A sign of a disease is: 1

(a) a food habit
(b) only what the patient feels
(c) something a doctor can observe or measure, such as a rash
(d) a medicine

8. The heating element of an electric heater is made of: 1

(a) plastic **(b)** copper **(c)** iron **(d)** nichrome

9. In an investigation of how the length of a pendulum affects its time period, the independent variable is: 1

(a) length of the pendulum **(b)** colour of the bob
(c) time period **(d)** mass of the room

10. Plant cells usually have a large central: 1

(a) nucleus only **(b)** vacuole **(c)** chloroplast only **(d)** membrane only

11. Scurvy is caused by deficiency of: 1

(a) vitamin C **(b)** iron **(c)** calcium **(d)** vitamin A

12. A zigzag line in a circuit diagram represents: 1

(a) a resistor **(b)** a bulb **(c)** a cell **(d)** a switch

13. In the same study, the time taken by the puri to puff up is the: 1

(a) independent variable **(b)** dependent variable
(c) hypothesis **(d)** control factor

14. The instrument that makes very small objects appear larger is the: 1

(a) microscope **(b)** periscope **(c)** telescope **(d)** stethoscope

15. Which of these is an example of social health? 1

(a) getting along well with family and neighbours
(b) having a normal body temperature
(c) being free from cough
(d) having a strong heart

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. What is a hypothesis? Give one example related to puri making. 2

17. Who discovered cells and how? Name the material he observed. 2

18. Define health as per the World Health Organization. 2

19. What is an electric circuit? 2

20. Why do scientists repeat an experiment several times? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Describe how you would plan an investigation to find the effect of oil temperature on the time taken by a puri to puff up. 3

22. Describe how you would prepare a slide of an onion peel and observe it. 3

23. Explain the three aspects of health with one example of each. 3

24. Draw in words a circuit with a cell, bulb and switch and explain how the bulb glows. 3

OR

Why is the scientific method not a rigid list of steps? Explain with a situation. 3

25. Why is the cell called the structural and functional unit of life? 3

OR

Name the deficiency diseases caused by lack of vitamin A, vitamin C and iron, giving one symptom of each. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Design a complete investigation to study whether the amount of water used in kneading dough affects how a puri puffs. Mention question, hypothesis, variables, constants, procedure and recording. 5

OR

Draw in words and label the parts of a typical plant cell and state the function of each. 5

27. Explain the concept of health as defined by WHO. Describe the physical, mental and social aspects with examples. 5

OR

Describe Oersted's experiment with a labelled set-up and give its conclusion. State the effect of reversing the current. 5

END OF QUESTION PAPER Total: 50 Marks

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